DP Biology – A4.2

Conservation of biodiversity

- What is driving the sixth mass extinction?
- What can conservationists do to reduce the loss of biodiversity?

SL statements A4.2.1–8. Textbook pp. 156–180.
Image: tropical forest in Thailand. Vyacheslav Argenberg, CC BY 4.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A4.2.1–3	Describe biodiversity at three levels, compare it with the past, and use case studies of human-caused extinction
2	A4.2.4–6	Explain causes of ecosystem loss, evaluate evidence of a crisis, and outline its main causes
3	A4.2.7–8	Compare approaches to conservation, and explain how the EDGE programme chooses species

The big picture: a sixth mass extinction?



Life is more diverse now than ever, yet species are being lost far faster than the **background rate**. Humans are the main cause.

Big question
With limited money, which species should we save first?

Biodiversity at three levels

A4.2.1

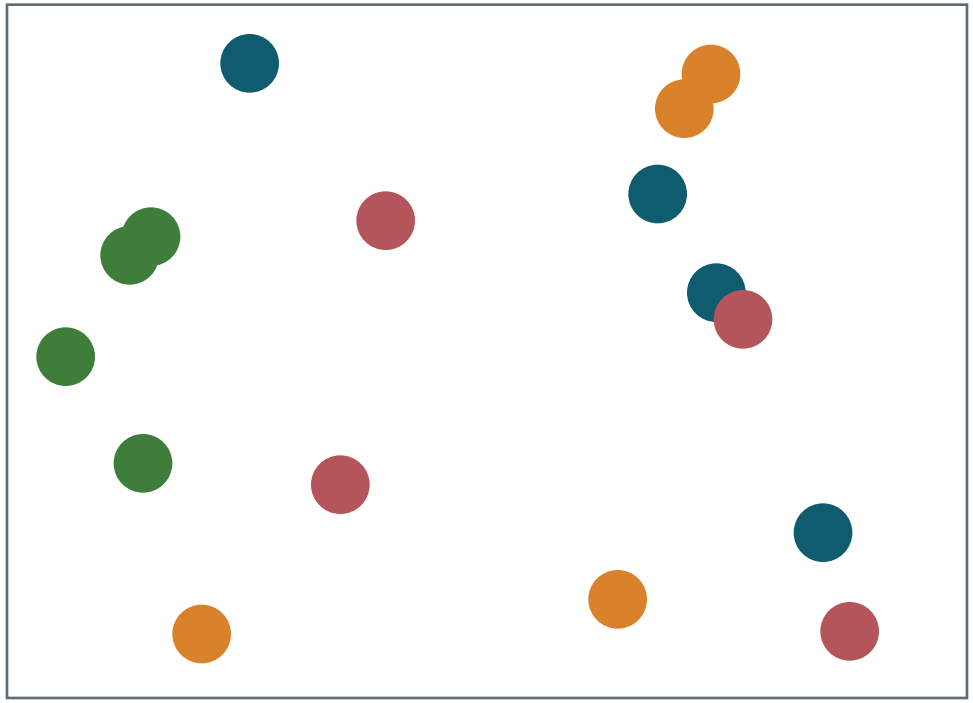
Ecosystem	Species	Genetic
Range of habitats in an area	Variety and abundance of species	Range of alleles within a species

Richness and evenness

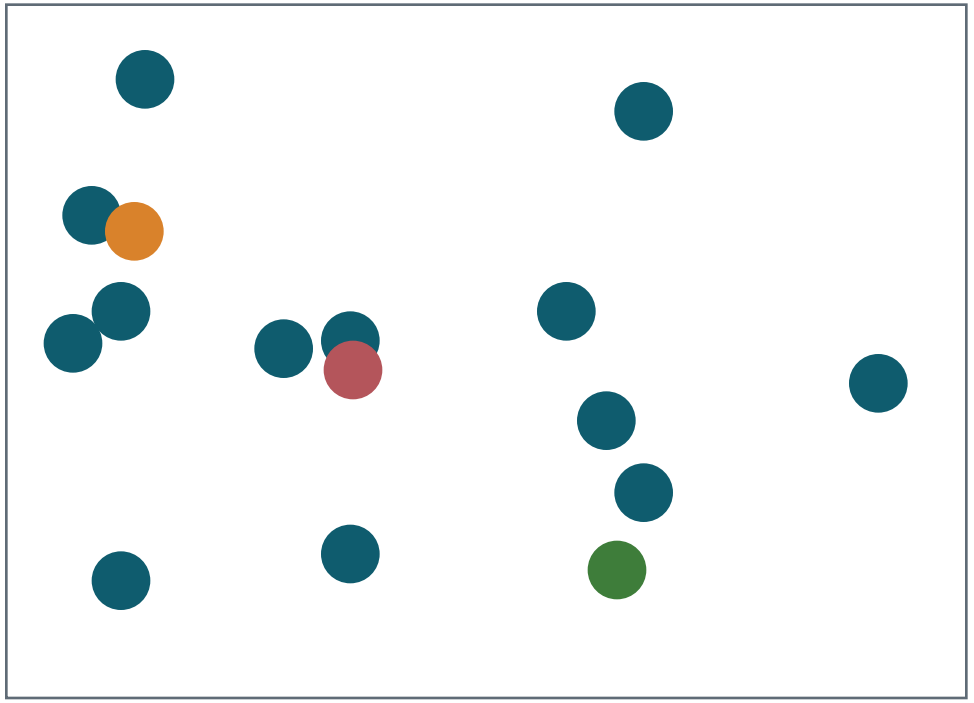
Richness: number of species. **Evenness**: how similar their numbers are. Same richness, but one dominant species means lower diversity. Why does that matter if disease strikes it?

Same richness, different diversity

A4.2.1



X: 4 species, even



Y: 4 species, one dominant

Key idea

X has higher diversity: same richness, greater evenness.

Simpson’s reciprocal index

A4.2.1

$$D = N(N - 1) / \text{sum of } n(n - 1)$$

N = total organisms; n = number of each species. Higher D = higher diversity.

Species	X	Y
A	10	37
B	10	1
C	10	1
D	10	1

Worked: X

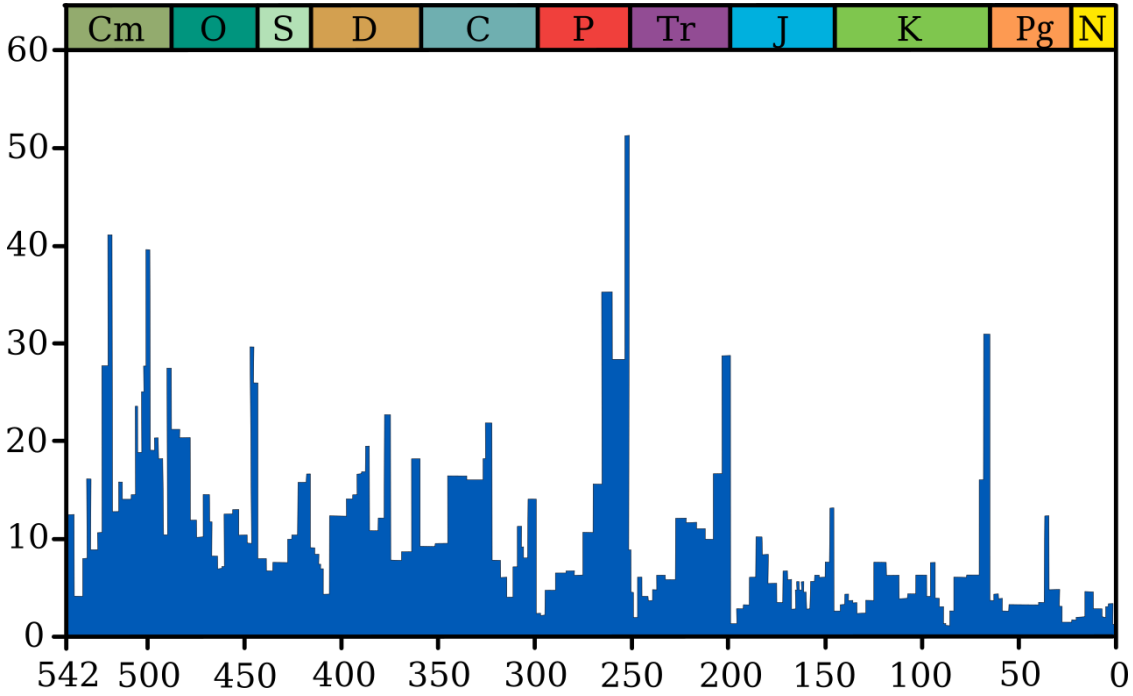
N = 40. Sum n(n-1) = 4 × 90 = 360. D = 40 × 39 / 360 = 4.33.

Your turn

Calculate D for Y and compare.

How does today compare with the past?

A4.2.2



Extinction intensity over time. CC BY-SA 3.0.

Fossils suggest **more species** are alive now than ever.

There have been **five mass extinctions**; diversity recovered after each.

Caution

Fossil records are incomplete. High diversity today does not mean all is well: extinction rates are very high.

Case study: moa in New Zealand

A4.2.3



Haast's eagle attacking moa. John Megahan, CC BY 2.5.

Nine species of flightless bird, up to about 3 m tall, evolved with few mammal predators.

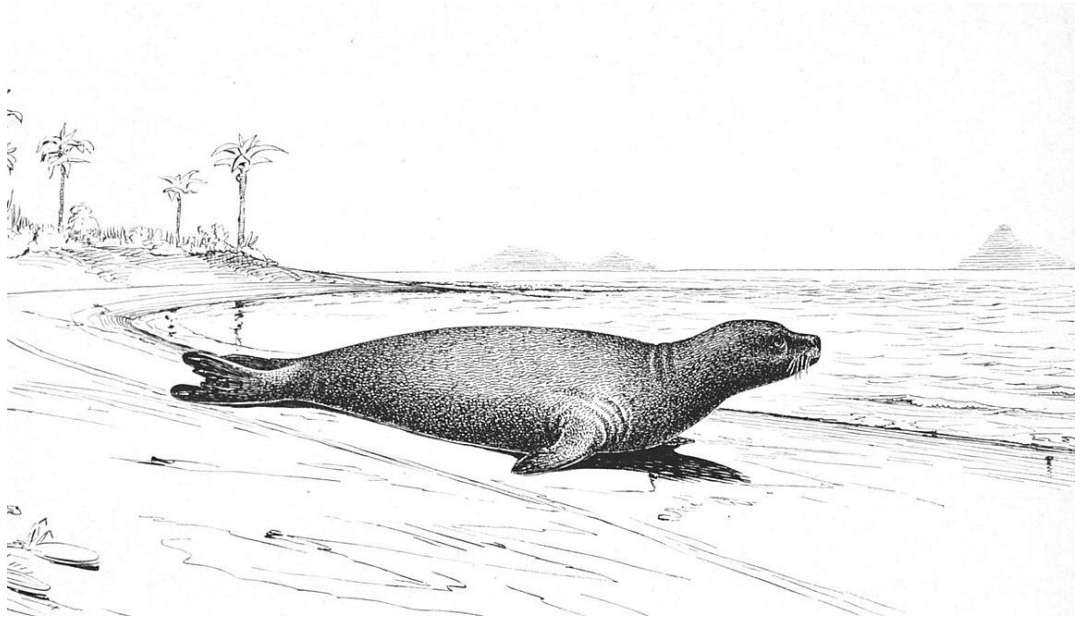
Humans arrived about 1280, hunted moa and burnt forest; giant moa were gone by about 1500, and Haast's eagle with them.

Key idea

Hunting plus habitat loss; species that evolved without such predators are most at risk.

Case study: the Caribbean monk seal

A4.2.3



Caribbean monk seal. Public domain.

Hunted for oil and meat; showed little fear of people.

Last seen 1952; declared extinct 2008.

Local example

Partridge numbers in Switzerland fell after farmland changed. Find a local case and its causes.

Check-in: class 1

A4.2.1–3

1. Name the three levels of biodiversity.
2. Distinguish richness and evenness.
3. State what a higher Simpson’s index means.
4. Explain why fossil comparisons are uncertain.
5. Outline the causes of moa extinction.

How to use this

Answer without your notes, then check against slides 3–9.

Causes of ecosystem loss

A4.2.4

Land use

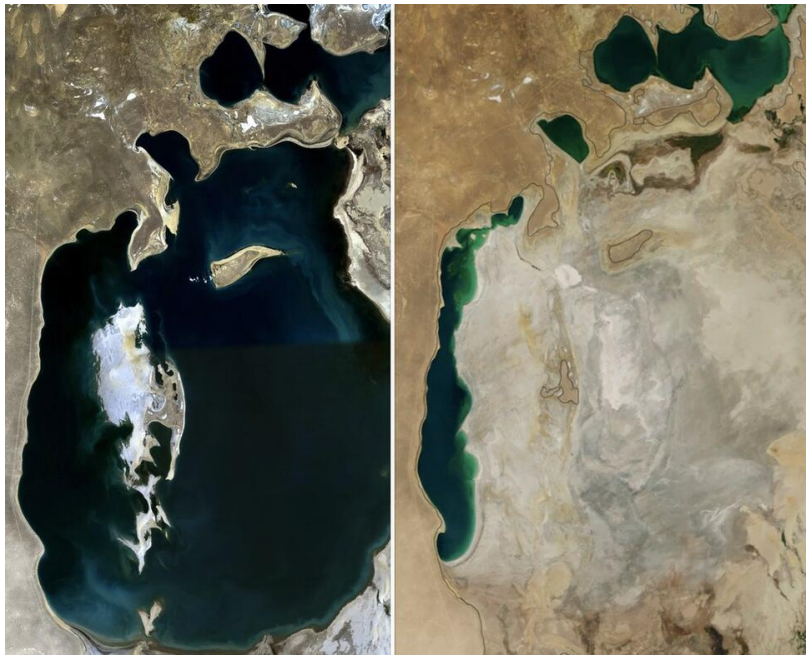
Farming, logging, mining, urban growth

Water

Dams, draining, diversion

Change

Fertilizer, pollution, climate change



Example: the Aral Sea

Rivers diverted for cotton: the lake shrank, salt rose, native fish died out and fishing collapsed. Image: 1989 and 2014, NASA, public domain.

Case study: mixed dipterocarp forest

A4.2.4



Tropical forest in Thailand. V. Argenberg, CC BY 4.0.

South-East Asian rainforest with huge tree and animal diversity.

Cleared for timber and **oil palm**; small fragments cannot support some species, and stored carbon is released.

Research

How has forest loss affected one named animal in Borneo?

Why fragments are not enough

A4.2.4



continuous forest

fragments: same total? small patches, edges

Key idea

Large animals need big ranges; edges are drier and hotter; patches isolate populations.

Local case study: ecosystems in Switzerland

A4.2.4



Peat moss in a bog. NC Wetlands, CC0.

Ecosystem	Decline	Why it matters
Moorland, bog	Drained; peat cut	Habitat; carbon store
Floodplain	Rivers straightened	Flood control

Evidence

Swiss moors have declined by almost 90%.

Research

Choose a local ecosystem: causes of loss and consequences.

Causes of the current crisis

A4.2.6

Overarching cause

A growing human population needs more land, food and resources.

Cause	Example
Over-exploitation	North Sea cod; moa
Urbanization	Land lost to buildings and roads
Deforestation	Clearing for farming and grazing
Pollution	Pesticides and plastics
Invasive species	Species carried to new places

Evidence of a biodiversity crisis

A4.2.5

IPBES estimates about a million species are threatened. Evidence: population sizes, ranges, richness, evenness and ecosystem degradation, across many habitats and groups.

Citizen science

Burying tea bags to measure decomposition; reporting the invasive beetle Popillia japonica; recording flowering times in an app.

Evaluate

Strengths and weaknesses of citizen data versus expert surveys: amount, spread, reliability.

Working with data: two plots

A4.2.1

Trees of each species in two plots. Invented values for practice.

Species	Plot A	Plot B
Oak	12	30
Ash	8	4
Beech	10	3
Birch	10	3

1. Calculate D for each plot.
2. Explain the difference.
3. Which plot better resists an oak disease?

Check-in: class 2

A4.2.4–6

1. Give three causes of ecosystem loss.
2. Outline the Aral Sea case.
3. Explain why fragments support fewer species.
4. State the overarching cause of the crisis.
5. Evaluate citizen science data.

How to use this

Answer without your notes, then check against slides 11–17.

Two ways to conserve

A4.2.7

In situ: in the natural habitat

Nature reserves, rewilding, restoring degraded land

Ex situ: outside the habitat

Zoos, botanic gardens, seed banks, frozen storage

Key idea

No single approach is enough: different species and habitats need different measures.

Why several approaches are needed

A4.2.7

	In situ	Ex situ
Meaning	Within natural habitats	Outside natural habitats
Examples	Reserves, rewilding	Zoos, seed banks
Strength	Protects whole ecosystems	Saves critically rare species
Weakness	Needs land and enforcement	Small gene pools; behaviour lost

Key idea

Habitat protection is usually most important; ex situ is a safety net.

Conservation near us: Switzerland

A4.2.7



Swiss National Park. H. Krapf, CC BY-SA 3.0.

In situ	Ex situ
Swiss National Park (1914)	Agroscope crop gene bank
UNESCO Biosphere Entlebuch	Zoo Basel breeding
Regional nature parks	Aubonne arboretum



European bison. Michael Gabler, CC BY 3.0.

Reintroduction

European bison survived only in captivity; zoo-bred animals now roam parts of Europe, with a trial herd in Switzerland.

Seed banks: a frozen safety net

A4.2.7



Strength

Huge genetic diversity in a small space, safe for decades or centuries.

Weakness

Seeds do not evolve with the climate; some species' seeds cannot be dried or frozen.

The EDGE of Existence programme

A4.2.8



Tree pangolin. Charles J. Sharp, CC BY-SA 4.0.

Evolutionarily Distinct

Few close relatives: much unique history lost if extinct

Globally Endangered

At high risk of extinction

Big question

Money is limited: EDGE scores help set priorities. Should distinct, little-known species come before tigers?

Working with data: choosing priorities

A4.2.8

Species	Distinctness	Threat
Aye-aye	High	Endangered
Tiger	Low	Endangered
Pangolin	High	Critically endangered
House sparrow	Low	Least concern

1. Rank the four for EDGE priority.
2. Explain your top choice.
3. Give one argument for funding the tiger anyway.

Check-in: class 3

A4.2.7–8

1. Distinguish in situ and ex situ conservation.
2. Give one strength and one weakness of each.
3. Explain why several approaches are needed.
4. State what EDGE stands for.
5. Explain why EDGE favours distinct species.

How to use this

Answer without your notes, then check against slides 19–24.

Common mistakes to avoid

Not this	But this
Biodiversity is just species number.	It includes genetic and ecosystem diversity and evenness.
Extinction is only a modern problem.	Humans have caused extinctions for millennia.
Zoos alone can save species.	Habitat protection usually matters more.

Summary: conservation of biodiversity

A4.2.1–8

	Key idea
A4.2.1	Three levels; richness, evenness; Simpson's index
A4.2.2	More species now than ever; five past mass extinctions
A4.2.3	Moa and monk seal
A4.2.4	Ecosystem loss: Aral Sea, dipterocarp forest
A4.2.5–6	Evidence and causes of the crisis
A4.2.7	In situ and ex situ
A4.2.8	EDGE priorities

Check that you can...

1. **Outline** three levels of biodiversity
2. **Calculate** Simpson's index
3. **Discuss** causes of extinction
4. **Evaluate** evidence of a crisis
5. **Compare** in situ and ex situ conservation
6. **Explain** EDGE priorities

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between richness and evenness.
2. State three causes of today's biodiversity crisis.
3. Explain why the EDGE programme might choose a species that has few close relatives.

Image credits

Virgin tropical forest in southern Thailand, Khlong Phanom, Surat Thani.jpg – Vyacheslav Argenberg – CC BY 4.0

Extinction intensity.svg – Author not stated – CC BY-SA 3.0

Giant Haasts eagle attacking New Zealand moa.jpg – John Megahan – CC BY 2.5

Monachus tropicalis.jpg – Author not stated – Public domain

AralSea1989 2014.jpg – NASA. Collage by Producercunningham. – Public domain

Sphagnum moss in Panthertown Valley bog.jpg – NC Wetlands from Raleigh, NC – CC0

Swiss National Park 045.JPG – Hansueli Krapf – CC BY-SA 3.0

Bison bonasus (Linnaeus 1758).jpg – Michael Gäbler – CC BY 3.0

Tree pangolin (Phataginus tricuspis) Nyamebe Bepo 5.jpg – Charles J. Sharp – CC BY-SA 4.0

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